Trishit Mukherjee

Mathematically grounded deep learning for inverse imaging and stable reconstruction

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Research Profile

Junior Research Fellow at IISc developing neural denoisers and learned regularizers with explicit convergence guarantees for iterative image reconstruction, with GPU implementations in PyTorch, C++, and CUDA. Seeking a PhD in inverse problems, computational imaging, theoretically grounded computer vision, and graph signal processing.

Education

M.Tech. in Artificial Intelligence, Indian Institute of Science, Bengaluru 2023–2025
CGPA: 8.4/10

B.E. in Computer Science and Engineering, Jadavpur University, Kolkata 2019–2023
CGPA: 8.81/10

Research Experience and Publications

Contractive Neural Networks for Stable Image Reconstruction *May 2024–Jun 2025*
Master's Thesis · Prof. Kunal N. Chaudhury, IISc
Stabilizing Deep Reconstruction Operators with Contractive Anchoring *ECCV 2026*. [arXiv | code]
Arghya Sinha, Trishit Mukherjee, and Kunal N. Chaudhury.
ViSTA: Viscosity Stabilized Plug-and-Play Reconstruction *2025*. [arXiv]
Arghya Sinha, Trishit Mukherjee, and Kunal N. Chaudhury.

- Formalized peak-and-collapse instability via a local stability index and developed adaptive contractive anchoring to stabilize black-box PnP/RED operators without retraining the pretrained denoiser.
- Designed contractive gradient-step CNN anchors using spectral constraints and learnable splines; validated on deblurring, superresolution, and MRI, sustaining 0.5–1 dB PSNR gains across iterations.

Trainable Kernel Denoisers for Convergent Image Reconstruction *Dec 2025–Present*
Prof. Kunal N. Chaudhury, IISc
Trainable Nonexpansive Denoisers for Contractive Image Reconstruction *ICML 2026*. [arXiv | code]
Arghya Sinha, Aditya Banerjee, Trishit Mukherjee, and Kunal N. Chaudhury.

- Proposed permutation-based weighted aggregation with global nonexpansiveness by construction; proved PnP-HQS contractivity under mild conditions and matched DRUNet denoising performance while ensuring stable deblurring and superresolution.

`neural_shift_cuda`: Parallel Shift Aggregation *Jul 2026*
Open-source C++/CUDA extension · Lead developer
[code]

- Fused serial shift aggregation into parallel CUDA kernels with full autograd, a PyTorch CPU fallback, and reference-equivalence tests, preserving the original models and operators.
- Achieved $2.23\times/1.31\times$ full-step speedups for improved symmetric/asymmetric kernel denoisers and $1.61\times/1.33\times$ for the original NECTR implementations; supports Pascal–Blackwell GPUs.

Location Metadata Extraction from Geosocial Road-Accident Data *Aug 2022–May 2023*
Research Project · Prof. Sarbani Roy, Jadavpur University
Location Metadata Extraction from Geosocial Data of Road Accidents using Deep Learning Models *Evolving Systems*, 16(1), 2025. [report]
Trishit Mukherjee, Soumitra Sinhahajari, Debargha Mukherjee, Hrishikesh Mallick, Asif Iqbal Middya, and Sarbani Roy.

- Built an API-based news pipeline with LSTM/Word2Vec relevance classification (96% accuracy, 0.94 F1) and BERT location extraction (0.673 exact match).

Work Experience and Teaching

Junior Research Fellow · LISA, EE, IISc *Aug 2025–Present*
Supervisor: Prof. Kunal N. Chaudhury · Bengaluru, India
Developing stochastic denoisers and graph/kernel-Laplacian regularizers using learned edge weights, primal-dual reconstruction, conjugate-gradient solves, and implicit differentiation.

Teaching Assistant · Linear Algebra and Applications, IISc *Aug–Nov 2024*
Instructor: Prof. Kunal N. Chaudhury · Bengaluru, India
Ran tutorials and helped develop assignments for undergraduate and graduate students.

Software Engineering Intern · Innovation Junction LLC *Jun–Aug 2021*
Remote, USA
Built AWS data pipelines for CSV operations, SQL/NoSQL ingestion into EC2, and Power BI reporting.